SIMATS ENGINEERING

ASSESSMENT TOOL 1: Analytical Problem Solving

COURSE NAME:	Cloud Computing and Big Data Analytics	**COURSE CODE:**	CSA15

COURSE OUTCOME COVERED

CO2: *Analyze virtualization architectures to identify performance and security issues in cloud computing environments.* (BL3)

Total Marks: 50

Dave's Taxonomy: Manipulation (Level 2)
Question Count: 5

Code of Conduct

I R.Ram Nikesh (192511302) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: ____________

Assessment Tasks

Analytical Problem Solving

1. Compare Type-1 and Type-2 hypervisors for a cloud datacenter hosting mission-critical applications. Analyze performance, security, and manageability, then justify the better choice.
2. A cloud provider must isolate workloads belonging to finance, healthcare, and student portals on the same hardware. Analyze how virtualization architecture supports isolation and identify two major attack surfaces.
3. Study the following VM host utilization snapshot and recommend corrective actions: CPU 92%, memory 88%, storage I/O latency 35 ms, average VM boot time 170 s. Explain the likely bottlenecks.
4. Analyze how Software Defined Datacenter architecture improves agility compared with a traditional datacenter. Support your answer with compute, storage, and network virtualization considerations.
5. A multi-cloud federation project fails due to identity mismatches between providers. Analyze the issue and propose a secure identity and privacy design using federation concepts.

Analytical Problem Solving Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 – Proficient	Level 2 - Developing	Level 1 - Beginning
Problem Interpretation	20%	Interprets all technical requirements accurately	Interprets most requirements correctly	Partial interpretation	Misinterprets the problem
Analytical Reasoning	25%	Strong reasoning with clear cause-effect analysis	Good reasoning with minor gaps	Basic analysis only	Weak or no analysis
Method Selection	20%	Chooses highly appropriate virtualization and security concepts	Chooses mostly suitable concepts	Partially suitable concepts	Inappropriate approach
Solution Quality	20%	Provides feasible and technically sound solution	Reasonable solution with minor issues	Basic solution with limitations	Unclear or invalid solution
Presentation of Analysis	15%	Well-structured and technically precise	Generally clear	Somewhat unclear	Difficult to follow

Total Marks Awarded:

Isolation techniques include:

* Separate Virtual CPUs
* Dedicated Virtual memory
* Virtual disks
* Virtual network interfaces

Each VM cannot directly access another VM's resources.

Benefits

* Data Confidentiality
* Better resource utilization
* Independent Software environments
* Fault isolation

Two Major Attack Surface

1. Hypervisor attacks

Attackers exploit vulnerabilities in hypervisor.

Eg: VM Escape attack, Attacker gains access from one VM to host system.

Impact: Complete compromise of all VMs

Protection:
* Regular Patching
* Secure Configuration
* Access Control

2. Virtual Network attacks

VMs communicate through Virtual Switches

Possible attacks

* Packet Sniffing
* ARP Spoofing
* VLAN hopping

Justification

For mission-critical cloud application, Type hypervisor is the better choice because it offers:

* Higher performance
* Better security
* Greater Scalabibity
* Enterprise - level management features

Conclusion

Type-1 hypervisors provide the reliability & security required for banking, healthcare, government & enterprise Cloud data Centers

2 A cloud Provider Must Isolate workloads belonging to Finance, Healthcare and Student Portal on the Same Hardware. Analyze how virtulization architecture Supports Isolation & identify two major attack Surfaces
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Introdution

* Virtualization allows multiple OS to run independently on the Same physical Server while maintaining complete isolation

Workload Isolation

Each department receives its own VM.

Physical Server

Hypervisor

├── Finance VM
├── Healthcare VM
└── Student Portal VM

Effects : * Memory Swapping
* Slow application Performance
* VM instability

Recommendation :
* Increase RAM
* Use memory Optimization
* Remove unused VMs

Storage bottleneck

35 ms latency is high

Normal Cloud Storage latency :
* SSD : 1-5ms
* HDD : 10-20ms

Effects : * Slow VM startup
* Delayed file access
* Poor database performance

Recommendation :
* Upgrade to SSD / NVMe
* Use storge caching
* optimize Storage I/O

Boot Time

170 seconds is unusually high

Reasons :
* Cpu overload
* Memory shortage
* Storage latency

Recommendation :
* Reduce Startup Services
* Optimize VM images
* Improve Storage ~~Peer~~ Performance

Protection:

* VLAN isolation
* Virtual firewalls
* Network Segmentation

Conclusion.

Virtualization architecture securely isolates multiple workloads while efficient security controls protect against hypervisor and virtual network attacks

3 Study the following VM host utilization snapshot and Recommend corrective Actions: CPU 92%, Memory 88%, storage I/O latency 35 ms, Avg VM boost time 170s Explain the likely bottlenecks.

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CPU utilization = 92%
Memory Usage = 88%
Storage latency = 35 ms
VM Boot Time = 170 seconds

Analysis

* CPU utilization above 90% indicate Processor overload.

Effects:

Slow VM execution
Increased response time
Scheduling delays

Recommendation

Add more CPU cores
Enable Load balancing
Migrate VMs

Memory bottleneck

88% memory utilization indicates insufficient RAM

Storage Virtualization

Multiple Storage devices are combined into one logical Storage Pool.

Benefits : Faster ~~develo~~ deployment

Storage Virtualization

Multiple Storage devices are Combined into one logical Storage Pool.

Benefits :
* Better utilization
* Dynamic allocation
* Easy backup

Network Virtualization

Uses Software defined networking (SDN)

Benefits :
* Programmable networking
* Virtual Switches
* Faster Configuration
* Network automation

Advantages of SDDC

* Faster provisioning
* Reduced operational cost
* High Scalability
* centralized management
* Improved disaster recovery

Conclusion

* Compared with traditional datacenters, SDDCs Provide greater agility, flexibility & automation through compute, Storage & network virtualization

Conclusion

The main bottlenecks are:

* CPU overload
* High memory usage
* Slow storage Subsystem

Improving these resources will Significantly reduce VM bo time & increase cloud performance

4 Analyze how Software defined datacenter (SDDC) Architectu Improves agility Compared with a traditional data cent Support your Answer with compute, storage & Network virtua -a Considerations

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Introduction

A Software defined datacenter (SDDC) virtualizes all onfrastructure Components and manages them through Softwars

Traditional

* Manual Server Setup
* physical networking
* dedicated storage devices
* Slow deployment

Software defined datacenter

* virtualized compute, Storage, network

Compute Virtualization

* Multiple VMs share hardware.
* Rapid VM Creation
* Automatic resource allocation
* Live migration.

Benefit :

* Faster deployment.

Multi-Cloud Federation Project Fails due to identity mismatches between Providers. Analyze the issue & Propose a secure identity and privacy design using federation concepts

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Introduction

Multi-cloud federation enables users to access multiple cloud providers using a common identity

Problem Analysis

Different cloud Providers use different identity Systems.

Examples: Azure Active directory
Google identity
AWS IAM

without federation:

Multiple usernames /Passwords
Authentication failures
Poor user experience

Proposed secure identity design

1. Identity Provider (idp)

* A trusted Identity Provider authenticates users once.

Examples: okta
keycloak
Azure AD

2. Signale Sign -on (SSO)

Users log in once & access multiple cloud platforms.

benefits:

* better user experience

* Reduced Password management

3. Federation Protocols

Use standard Protocols such as:

SAML
OAuth 2.0
OpenID connect

These enable secure authentication between Providers.

Privacy Protection

* Encrypt identity data during transmission
* Use Multi-Factor Authentication (MFA).
* Apply Role-Based Access control (RBAC)
* Share only nessary user attributs
* Maintein audit logs for monitoring & compliance

Benefits

* Secure authentication
* Better interoperability
* Improved privacy
* simplified user management
* Reduced identity mismatches

Conclusion

* A federated identity System using SSO, std federation Protocols, MFA, RBAC & encrypted Communication ensures secure & seamless access across multiple cloud Providers while Protecting user privacy.